

SMARTBRIDGE SKILL WALLET

Code-Layout, Readability And Reusability Evaluation Report

Project Name	edugenie google gemini powered learning assistant
Date	07/07/26
Team ID	XXXXXX
Maximum Marks	5 Marks

Evaluation Purpose: This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/ tabs used throughout the code	Yes	Standardized 4-space Python PEP 8 style formatting applied across all modules.
2	Proper File Structure	Files and folders are logically organized	Yes	Clear isolation of API wrappers, configuration, UI components, and utility directories.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive snake_case used contextually (e.g., gemini_response_payload, user_session_state).
4	Function/ Method Names	Functions are descriptively named	Yes	Precise verb-noun actions implemented (e.g., generate_personalized_quiz(), stream_llm_response()).

S.No	Code Quality Parameter	Description	Followed	Remarks
5	Code Comments	Inline and block comments explain logic	Yes	Comprehensive docstrings detailing parameters and returns are present in all service endpoints.
6	Modular Design	Code is split into reusable functions/modules	Yes	Core Google Gemini connection logic isolated entirely from the frontend layout scripts.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Redundant payload transformations refactored seamlessly into helper utility methods.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Robust try-except structures capture API rate-limiting and connection issues gracefully.

Reusable Components / Modules

S.No	Component/Module Name	Language / Technology	Where Reused	Reusability Level
1	GeminiClientManager	Python / google-genai SDK	Quiz Generator, Smart Summary, Chatbot UI	High
2	DocumentParsingEngine	Python / PyPDF2	File Upload, Notes Extractor Modules	High
3	SessionStateManager	Python / Streamlit	Multi-page Navigation, User Auth Check	High
4	MarkdownFormatter	Python / HTML	AI Response Rendering, Export to PDF Tool	Medium
5	ValidationSchema	Python / Pydantic	JSON structured output confirmation	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5.0	Perfect modular layout following modern software engineering directories.
Readability	4.8	Highly readable codebase due to strict conformance with variable naming standards.
Reusability	5.0	Exceptional design of standalone client modules ready to be ported into future microservices.
Documentation / Comments	4.5	Strong module level docstrings; could benefit from minor increase in inline edge-case comments.
Overall Score	4.8 / 5.0	Superb repository health indicating an easily maintainable and extensible AI application product.